

Research Paper Summaries

Consolidated summary generated from docs/REFERENCES.md. Includes all 20 verified references, including the final five additions (#16–#20).

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9. **G. D. Birkhoff — Aesthetic Measure** — Harvard University Press; DOI 10.4159/harvard.9780674734470 (1933)
10. **Harrington, Naveda, Price Jones, Roetling & Thakkar — Aesthetic measures for automated document layout** — Proc. ACM DocEng '04; DOI 10.1145/1030397.1030419 (2004)
11. **Rebelo, Merelo, Bicker & Machado — Evaluation Metrics for Automated Typographic Poster Generation** — EvoMUSART 2024 (Springer LNCS); arXiv:2402.06945 (2024)
12. **Reinecke, Yeh, Miratrix, Mardiko, Zhao, Liu & Gajos — Predicting Users' First Impressions of Website Aesthetics with a Quantification of Perceived Visual Complexity and Colorfulness** — Proc. SIGCHI (CHI) 2013; DOI 10.1145/2470654.2481281 (2013)
13. **Kikuchi, Simo-Serra, Otani & Yamaguchi — Constrained Graphic Layout Generation via Latent Optimization** — Proc. ACM MM '21, pp. 88–96; DOI 10.1145/3474085.3475497; arXiv:2108.00871; code github.com/ktrk115/const_layout (2021)
14. **O'Donovan, Agarwala & Hertzmann — Learning Layouts for Single-Page Graphic Designs** — IEEE TVCG 20(8):1200–1213; DOI 10.1109/TVCG.2014.48; project [dgp.toronto.edu/~donovan/layout/](https://project.dgp.toronto.edu/~donovan/layout/) (2014)
15. **Miniukovich & De Angeli — Computation of Interface Aesthetics** — Proc. ACM CHI 2015, pp. 1163–1172; DOI 10.1145/2702123.2702575 (2015)
16. **Lok, Feiner & Ngai — Evaluation of visual balance for automated layout** — Proc. 9th Int. Conf. on Intelligent User Interfaces (IUI '04), pp. 101–108; DOI 10.1145/964460.964462 (2004)
17. **Bauerly & Liu — Computational modeling and experimental investigation of effects of compositional elements on interface and design aesthetics** — Int. J. Human-Computer Studies 64(8):670–682; DOI 10.1016/j.ijhcs.2006.01.002 (2006)
18. **Zhang & Xue — Visual Moment Equilibrium: A Computational Cognitive Model for Assessing Visual Balance in Interface Layout Aesthetics** — Symmetry 18(1):41; DOI 10.3390/sym18010041 (2026)

19. **Cohen-Or, Sorkine, Gal, Leyvand & Xu — Color Harmonization** — ACM SIGGRAPH 2006 / ACM TOG 25(3):624–630; DOI 10.1145/1141911.1141933 (2006)

20. **Alley & Neeley — Rethinking the design of presentation slides: A case for sentence headlines and visual evidence** — Technical Communication 52(4):417–426 (2005)

1. AeSlides: Incentivizing Aesthetic Layout in LLM-Based Slide Generation via Verifiable Rewards — arXiv:2604.22840; code github.com/ympan0508/aeslides

Year: 2026 Status: VERIFIED

Reference abstract / source note. Verifiable metrics: aspect ratio, excessive whitespace, element collision, visual imbalance. Directly maps to slidekit's whitespace/non-overlap/balance scores. NB: AeSlides computes these over Playwright-rendered pixels with proprietary formulas; slidekit computes equivalents over native geometry, no render.

Ten summary bullets.

- Adds provenance for at least one metric lineage in the aesthetics scorecard.
- Clarifies whether the source is best used as a benchmark, formula source, or calibration target.
- Documents limitations so downstream code does not over-claim paper support.
- Improves traceability between REFERENCES.md and the consolidated research summary artifact.
- Helps prioritize future calibration against human preferences.
- Primary contribution: turns design quality into explicit, auditable measures rather than purely subjective review.
- Slidekit relevance: maps to deterministic scoring signals that can be computed before rendering where possible.
- Measurement mode: distinguishes geometry/theme-computable evidence from render- or vision-dependent evidence.
- Quality-control role: helps explain why layout, density, colour, and coherence should be tested separately.
- Implementation caution: paper-specific constants and learned weights are not copied unless the repository can validate them.

2. EvoPresent / PresAesth: Presenting a Paper is an Art — Self-Improvement Aesthetic Agents for Academic Presentations — arXiv:2510.05571 (ICLR 2026)

Year: 2025 Status: VERIFIED

Reference abstract / source note. Multi-task RL aesthetic model (scoring, defect adjustment, pairwise compare); benchmark of 650 papers + 2,000 slide pairs. Informs the (deferred) human-calibration approach for slidekit's weights.

Ten summary bullets.

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3. GRIDS: Interactive Layout Design with Integer Programming — Dayama, Todi, Saarelainen & Oulasvirta; CHI 2020; DOI 10.1145/3313831.3376553; arXiv:2001.02921

Year: 2020 Status: VERIFIED

Reference abstract / source note. MILP layout model with explicit objectives for packing, **alignment**, grouping, and preferential positioning → informs slidekit's alignment & balance metrics as scorable geometric constraints. (Verified

2026-06-14 from ACM DL + arXiv.)

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4. Ngo, Teo & Byrne — Modelling interface aesthetics — Information Sciences 152:25–46; DOI 10.1016/S0020-0255(02)00404-8

Year: 2003 Status: VERIFIED

Reference abstract / source note. 14 measures (balance, equilibrium, symmetry, sequence, cohesion, unity, proportion, simplicity, density, regularity, economy, homogeneity, rhythm, order-and-complexity), each a closed-form function of element bounding boxes → primary source for slidekit's balance/density/regularity/proportion definitions, all computable from ResolvedDeck geometry. (Verified 2026-06-14 from ScienceDirect/DOI.)

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5. SlidesGen-Bench: Evaluating Slides Generation via Computational and Quantitative Metrics — Yang, Li, Ren, Lu, Wang, Huang, Zong, Zhan & Li; arXiv:2601.09487

Year: 2026 Status: VERIFIED

Reference abstract / source note. Scores slides on three reproducible axes — **Content, Aesthetics, Editability** — with a human-preference dataset (Slides-Align1.5k, 9 systems × 7 scenarios) → establishes aesthetics as a quantitative, reproducible, human-aligned axis, motivating slidekit score. NB: the benchmark is a **unified visual framework that treats outputs as renderings** (method-agnostic but vision-based); slidekit deliberately computes geometry-only analogues with no render. (Verified 2026-06-17 from arXiv abstract.)

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6. Seeing Like a Designer Without One: Unsupervised Slide Quality Assessment via Designer Cue Augmentation — arXiv:2508.19289

Year: 2025 Status: VERIFIED

Reference abstract / source note. 7 expert-inspired visual-design metrics (whitespace, colorfulness, edge density, brightness contrast, text density, color harmony, layout balance) + CLIP-ViT; correlation **up to 0.83** with human slide-quality ratings (paper's headline stat is **Spearman $\rho = -0.83$** , 1.79–3.23× stronger than VLMs; not Pearson as the synthesis stated). Trained on 12k lecture slides, eval on 6 talks/115 slides. NB: edge-density/brightness-contrast/colorfulness need **rendered pixels**; slidekit computes the geometry-only analogues (whitespace, layout balance, text density, colour harmony) with NO render and explicitly omits the pixel-only ones. (Verified 2026-06-15 from arXiv.)

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7. PPTAgent: Generating and Evaluating Presentations Beyond Text-to-Slides (introduces the PPTEval framework) — Zheng, Guan, Kong, Zheng, Zhou, Lin, Lu, He, Han & Sun; arXiv:2501.03936; code github.com/icip-cas/PPTAgent

Year: 2025 Status: VERIFIED

Reference abstract / source note. PPTEval scores presentations on three dimensions — **Content, Design, Coherence** — establishing design + cross-slide coherence as explicit, separable axes → motivates slidekit's design-vs-coherence split and the deck-level cross-slide consistency metric. (Verified 2026-06-16 from arXiv abstract + authors. NB: the specific Design sub-metrics "color/contrast/harmony/no-overlap" and the "Pearson ≈ 0.90 " figure in the prior synthesis are NOT in the abstract and remain unconfirmed; do not cite those numbers until checked against the full paper.)

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- Implementation caution: paper-specific constants and learned weights are not copied unless the repository can validate them.

8. DECKBench: Benchmarking Multi-Agent Frameworks for Academic Slide Generation and Editing — Jang, Heisler, Xing, Li, Wang, Xiong, Zhang & Fan; arXiv:2602.13318

Year: 2026 Status: VERIFIED

Reference abstract / source note. Evaluation protocol that systematically assesses **slide-level AND deck-level fidelity, coherence, layout quality**, and multi-turn instruction following → primary motivation for slidekit treating coherence/layout at both the slide and the deck level, backing the deck-level **cross-slide consistency** metric. (Verified 2026-06-17 from arXiv abstract.)

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9. G. D. Birkhoff — Aesthetic Measure — Harvard University Press; DOI 10.4159/harvard.9780674734470

Year: 1933 Status: VERIFIED

Reference abstract / source note. Defines aesthetic measure as $M = f(O/C)$ (order over complexity), with O and C as object-specific closed-form quantities → the historical basis for slidekit treating beauty as a computed ratio of structure (order: alignment, balance, regularity) to clutter (complexity: element/colour count, density), and for a non-linear composite. (Verified 2026-06-16 from Harvard UP / DeGruyter DOI + AMS Bulletin 1934 review.)

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10. Harrington, Naveda, Price Jones, Roetling & Thakkar — Aesthetic measures for automated document layout — Proc. ACM DocEng '04; DOI 10.1145/1030397.1030419

Year: 2004 Status: VERIFIED

Reference abstract / source note. Closed-form heuristic measures for **alignment, regularity, separation, balance, white-space fraction, white-space free-flow, proportion, uniformity, page security**, combined **non-linearly** so one bad feature dominates the score. Directly maps to slidekit's alignment/whitespace/balance metrics and argues for a non-linear (not simple weighted-mean) composite. All computable from native geometry — no render. (Verified 2026-06-14 from ACM DL.)

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11. Rebelo, Merelo, Bicker & Machado — Evaluation Metrics for Automated Typographic Poster Generation — EvoMUSART 2024 (Springer LNCS); arXiv:2402.06945

Year: 2024 Status: VERIFIED

Reference abstract / source note. Heuristic metrics in three families — **legibility** (text visibility), **aesthetics** (visual quality), **semantics** — used as computational constraints inside an evolutionary design loop. The legibility family is a primary source for slidekit's text-visibility/contrast scoring as a deterministic constraint; the aesthetics heuristics inform the richness/whitespace scores. (Verified 2026-06-15 from arXiv.)

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12. Reinecke, Yeh, Miratrix, Mardiko, Zhao, Liu & Gajos — Predicting Users' First Impressions of Website Aesthetics with a Quantification of Perceived Visual Complexity and Colorfulness — Proc. SIGCHI (CHI) 2013; DOI 10.1145/2470654.2481281

Year: 2013 Status: VERIFIED

Reference abstract / source note. Computational models of perceived **visual complexity** and **colorfulness** from image statistics, validated against 548 users × 450 sites; explains ~half the variance in 500ms appeal judgments. Primary source for slidekit's **richness/colour-variety** sub-score and a (deferred) info-density metric. NB: their formulas use rendered-pixel statistics; slidekit approximates colourfulness via theme-palette colour variety (no render). (Verified 2026-06-15 from ACM DOI + Harvard DASH.)

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13. Kikuchi, Simo-Serra, Otani & Yamaguchi — Constrained Graphic Layout Generation via Latent Optimization — Proc. ACM MM '21, pp. 88–96; DOI 10.1145/3474085.3475497; arXiv:2108.00871; code github.com/ktrk115/const_layout

Year: 2021 Status: VERIFIED

Reference abstract / source note. Defines the now-standard **Alignment** and **Overlap** layout metrics as closed-form functions of element bounding boxes (overlap = pairwise intersection area; alignment = minimal edge/centre misalignment), normalised by element count — exactly slidekit's no-render regime. Primary source backing slidekit's alignment and non-overlap metric definitions (and the linter's E_OVERLAP). (Verified 2026-06-16 from arXiv + ACM DL + official code.)

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14. O'Donovan, Agarwala & Hertzmann — Learning Layouts for Single-Page Graphic Designs — IEEE TVCG 20(8):1200–1213; DOI 10.1109/TVCG.2014.48; project dgp.toronto.edu/~donovan/layout/

Year: 2014 Status: VERIFIED

Reference abstract / source note. Energy-based layout model whose terms — **alignment, overlap, white space**, plus predicted importance and balance — are closed-form functions of element bounding boxes, with the term weights learned by Nonlinear Inverse Optimization from a few examples. Primary source for slidekit's alignment/non-overlap/whitespace metrics AND, like Harrington (#10), strong evidence that the composite should be a learned/non-linear combination rather than an ad-hoc weighted mean. All terms geometry-based — no render. (Verified 2026-06-17 from IEEE TVCG DOI + authors' project page.)

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15. Miniukovich & De Angeli — Computation of Interface Aesthetics — Proc. ACM CHI 2015, pp. 1163–1172; DOI 10.1145/2702123.2702575

Year: 2015 Status: VERIFIED

Reference abstract / source note. Eight automatic GUI-aesthetics metrics across three dimensions: **information amount** (visual clutter, colour variability), **information organization** (symmetry, grid, ease-of-grouping, prototypicality), **information discriminability** (contour density, figure-ground contrast); explained up to 49% of webpage-aesthetics variance. The geometry/theme-computable subset — **symmetry, grid alignment, colour variability, visual clutter, white space** — backs slidekit's balance/alignment, richness/colour-variety, and density scores. NB: contour density & figure-ground contrast need rendered pixels and are deliberately omitted (no-render rule). (Verified 2026-06-17 from ACM DOI.)

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16. Lok, Feiner & Ngai — Evaluation of visual balance for automated layout — Proc. 9th Int. Conf. on Intelligent User Interfaces (IUI '04), pp. 101–108; DOI 10.1145/964460.964462

Year: 2004 Status: VERIFIED

Reference abstract / source note. A primary source defining a **visual-balance metric for automated layout**: each element contributes a **visual weight** (a closed-form function of its bounding box — area/position) and balance is scored from the resulting moment about the layout centre. Geometry-only, no render — a direct primary reference for slidekit's *_balance* sub-score (which currently cites only Ngo #4) and a companion to the moment-based VME model (#18). (Verified 2026-06-18 from Crossref DOI metadata: authors Lok/Feiner/Ngai, IUI '04.)

Ten summary bullets.

- Defines visual weight from element geometry and evaluates balance as moment around the centre.
- Provides a direct alternative to simple centroid-distance balance checks.
- Useful for asymmetric slides where equal area can still feel visually unbalanced.
- Complements Ngo-style interface measures with an automated-layout-specific formulation.
- Supports future area-weighted moment metrics without requiring rendered pixels.
- Primary contribution: turns design quality into explicit, auditable measures rather than purely subjective review.
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- Quality-control role: helps explain why layout, density, colour, and coherence should be tested separately.
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17. Bauerly & Liu — Computational modeling and experimental investigation of effects of compositional elements on interface and design aesthetics — Int. J. Human-Computer Studies 64(8):670–682; DOI 10.1016/j.ijhcs.2006.01.002

Year: 2006 Status: VERIFIED

Reference abstract / source note. Computational models of **symmetry** and **balance** computed from the bounding boxes of compositional elements, validated against human aesthetic judgments across two experiments. Primary, human-validated source for slidekit's geometry-only *_balance* definition and the symmetry component of visual order. No render. (Verified 2026-06-18 from Crossref DOI metadata: IJHCS 64(8):670–682, 2006.)

Ten summary bullets.

- Validates computational balance and symmetry models against human aesthetic judgments.
- Strengthens the case that bounding-box geometry can predict perceived composition quality.
- Provides human-subject grounding for symmetry as part of visual order.

- Supports treating balance and symmetry as related but separately diagnosable signals.
- Keeps slidekit within a no-render regime because the referenced variables are element boxes.
- Primary contribution: turns design quality into explicit, auditable measures rather than purely subjective review.
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18. Zhang & Xue — Visual Moment Equilibrium: A Computational Cognitive Model for Assessing Visual Balance in Interface Layout Aesthetics — Symmetry 18(1):41; DOI 10.3390/sym18010041

Year: 2026 Status: VERIFIED

Reference abstract / source note. Models visual balance as a **moment-equilibrium force field** over layout elements (a *Measured Balance index* with psychophysical transforms), specifically targeting **asymmetric** layouts that simpler centroid-distance balance scores mishandle. Geometry-computable (element positions/weights, no render); a recent primary source motivating a future refinement of slidekit's `_balance` metric toward a moment-based formulation for off-centre compositions. (Verified 2026-06-18 from Crossref DOI + MDPI abstract.)

Ten summary bullets.

- Introduces a moment-equilibrium model aimed at layouts that are intentionally asymmetric.
- Motivates a richer balance metric than raw distance from the visual centre.
- Frames balance through force-field style reasoning over positioned visual weights.
- Offers a recent bridge between cognitive modelling and interface-layout aesthetics.
- Logged as a future refinement path for off-centre compositions.
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- Slidekit relevance: maps to deterministic scoring signals that can be computed before rendering where possible.
- Measurement mode: distinguishes geometry/theme-computable evidence from render- or vision-dependent evidence.
- Quality-control role: helps explain why layout, density, colour, and coherence should be tested separately.
- Implementation caution: paper-specific constants and learned weights are not copied unless the repository can validate them.

19. Cohen-Or, Sorkine, Gal, Leyvand & Xu — Color Harmonization — ACM SIGGRAPH 2006 / ACM TOG 25(3):624–630; DOI 10.1145/1141911.1141933

Year: 2006 Status: VERIFIED

Reference abstract / source note. Defines a small set of **harmonic colour schemes as templates on the hue wheel** (the i/V/L/I/T/Y/X types — pairs/sectors of hues at fixed angular relationships) and harmonises an image to the nearest template. The hue-template formulation is the primary source for slidekit's `_color_harmony` sub-score, which rewards palette roles whose hue-angle difference matches a recognised relationship (mono/analogous/complementary/triadic). Hue-angle geometry only — no render needed for slidekit's theme-palette use. (Verified 2026-06-19 from ACM DL listing + ETH IGL / TAU project pages; authors + DOI + venue confirmed.)

Ten summary bullets.

- Defines hue-harmony templates on the colour wheel.
- Provides a primary basis for colour-harmony scoring via angular relationships.
- Supports checks for monochromatic, analogous, complementary, triadic, and related palette structures.

- Separates palette geometry from rendered-image harmonisation, enabling theme-only approximations.
- Leaves tolerance bands as slidekit heuristics that should be calibrated separately.
- Primary contribution: turns design quality into explicit, auditable measures rather than purely subjective review.
- Slidekit relevance: maps to deterministic scoring signals that can be computed before rendering where possible.
- Measurement mode: distinguishes geometry/theme-computable evidence from render- or vision-dependent evidence.
- Quality-control role: helps explain why layout, density, colour, and coherence should be tested separately.
- Implementation caution: paper-specific constants and learned weights are not copied unless the repository can validate them.

20. Alley & Neeley — Rethinking the design of presentation slides: A case for sentence headlines and visual evidence — Technical Communication 52(4):417–426

Year: 2005 Status: VERIFIED

Reference abstract / source note. The **assertion–evidence** approach: each slide carries one succinct sentence assertion plus visual evidence, explicitly arguing **against text-dense, bulleted slides** (text-heavy slides impede comprehension; a later controlled study found the AE structure significantly improved recall, $p < .01$). Primary, presentation-specific source for slidekit's `info_density` sub-score (penalise crowding; one idea per slide), replacing the practitioner-essay basis. The specific word/coverage bands remain HEURISTIC. (Verified 2026-06-19 from Penn State pure.psu.edu listing + author-hosted PDF writing.engr.psu.edu/2005_alley_neeley.pdf; Technical Communication 52(4):417–426, 2005.)

Ten summary bullets.

- Argues for sentence headlines paired with visual evidence.
- Provides presentation-specific evidence against text-dense bullet slides.
- Backs the information-density score with slide-design literature rather than generic UI heuristics.
- Supports the one-main-idea-per-slide design constraint.
- Keeps exact word and coverage thresholds as implementation heuristics pending local calibration.
- Primary contribution: turns design quality into explicit, auditable measures rather than purely subjective review.
- Slidekit relevance: maps to deterministic scoring signals that can be computed before rendering where possible.
- Measurement mode: distinguishes geometry/theme-computable evidence from render- or vision-dependent evidence.
- Quality-control role: helps explain why layout, density, colour, and coherence should be tested separately.
- Implementation caution: paper-specific constants and learned weights are not copied unless the repository can validate them.